Differentiation & Integration

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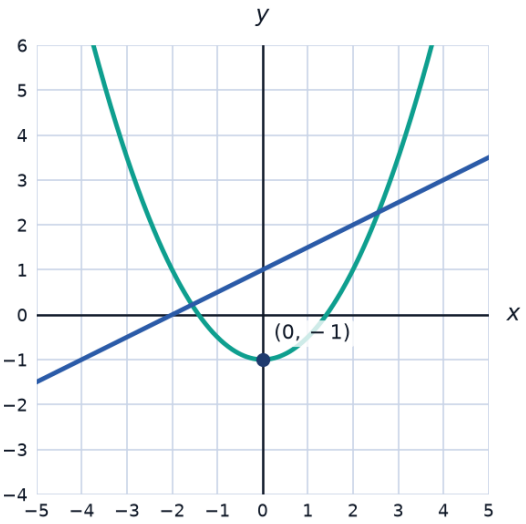
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Derivatives of Polynomials & Tangents

Find the x-coordinates of the stationary points of $y = x^3 + 15/2x^2 + 18x$.

2 [Foundational] • Target: Stationary Points & Optimization

Find the x-coordinates of the stationary points of $y = 2x^3 + 12x^2 + 18x$.



3 [Foundational] • Target: Indefinite & Definite Integrals

Find the x-coordinates of the stationary points of $y = 2x^3 - 9x^2 - 24x$.

4 [Foundational] • Target: Area Under Curves & Exam-Style Modeling

Find the x-coordinates of the stationary points of $y = 3x^3 + 9x^2 - 72x$.

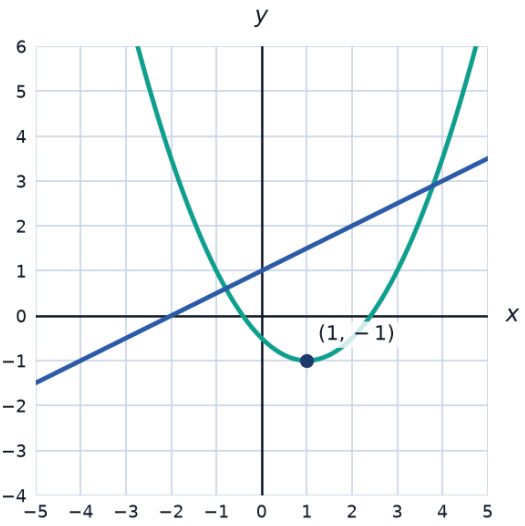
5 [Foundational] • Target: Derivatives of Polynomials & Tangents

Find the x-coordinates of the stationary points of $y = 3x^3 - 9/2x^2 - 54x$.

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6 [Foundational] • Target: Stationary Points & Optimization

Find the x-coordinates of the stationary points of $y = x^3 + 3/2x^2 - 18x$.



7 [Foundational] • Target: Indefinite & Definite Integrals

Find the x-coordinates of the stationary points of $y = x^3 - 3/2x^2 - 6x$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Area Under Curves & Exam-Style Modeling

Find the x-coordinates of the stationary points of $y = 2x^3 + 12x^2$.

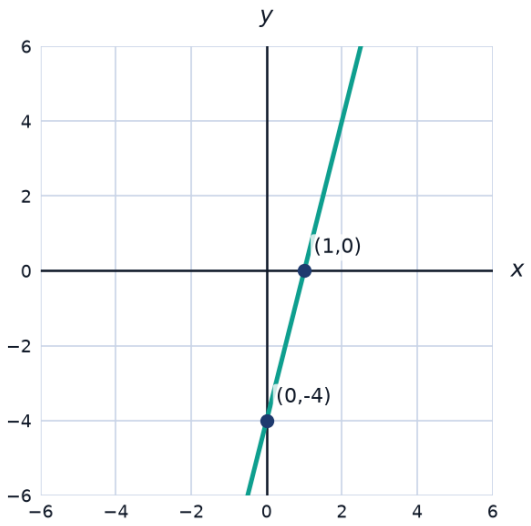
9 [Intermediate] • Target: Derivatives of Polynomials & Tangents

Find the x-coordinates of the stationary points of $y = x^3 - 12x$.

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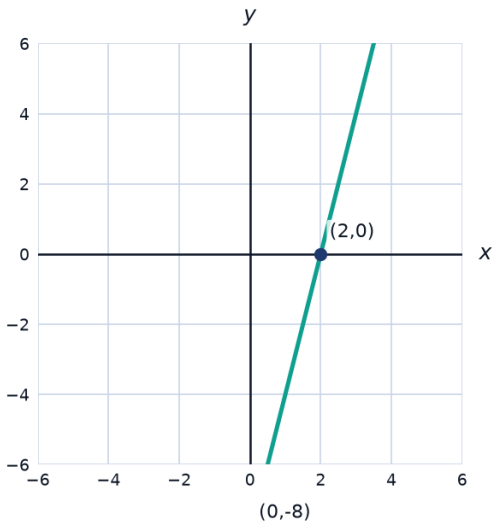
10 **[Intermediate]** • **Target: Stationary Points & Optimization**

Find the area enclosed between the curve $y = 4x - 4$, the x-axis, and the lines $x = 1$ and $x = 4$.



11 **[Intermediate]** • **Target: Indefinite & Definite Integrals**

Find the area enclosed between the curve $y = 4x - 8$, the x-axis, and the lines $x = 2$ and $x = 4$.



12 **[Intermediate]** • **Target: Area Under Curves & Exam-Style Modeling**

Find the area enclosed between the curve $y = 2x^2$, the x-axis, and the lines $x = 0$ and $x = 4$.

13 **[Intermediate]** • **Target: Derivatives of Polynomials & Tangents**

Find the area enclosed between the curve $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 4$.

14 **[Intermediate]** • **Target: Stationary Points & Optimization**

Find the area enclosed between the curve $y = 2x^2$, the x-axis, and the lines $x = 0$ and $x = 3$.

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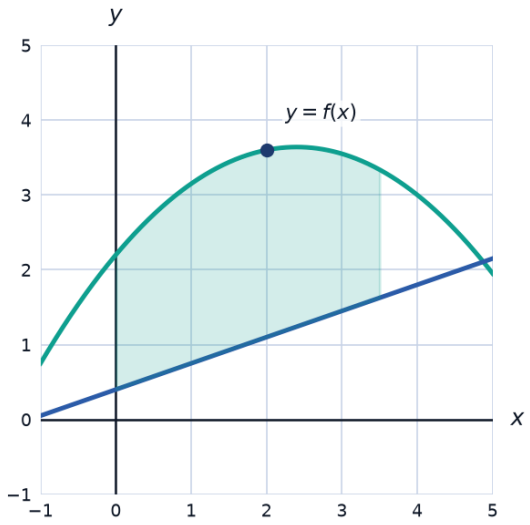
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Indefinite & Definite Integrals • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 3$.

16 [Challenging] • Target: Area Under Curves & Exam-Style Modeling • Cross-Lesson: 3.01 Graphs & Transformations

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 5$.



17 [Challenging] • Target: Derivatives of Polynomials & Tangents • Cross-Lesson: 4.01 Coordinate Geometry

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 5$.

18 [Challenging] • Target: Stationary Points & Optimization • Cross-Lesson: 5.01 Trigonometry

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 3$.

19 [Challenging] • Target: Indefinite & Definite Integrals • Cross-Lesson: 7.01 Statistics & Probability

Find the area enclosed between the curve $y = x$, the x-axis, and the lines $x = 0$ and $x = 3$.

20 [Challenging] • Target: Area Under Curves & Exam-Style Modeling • Cross-Lesson: 8.01 Functions: Composite & Inverse

Find the area enclosed between the curve $y = 3x^2$, the x-axis, and the lines $x = 0$ and $x = 2$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Derivatives of Polynomials & Tangents

Differentiate $y = 3x^4 - 5x^2 + 7x - 2$ with respect to x.

a. $12x^3 - 10x + 7$

b. $12x^3 - 10x^2 + 7$

c. $7x^3 - 10x + 7$

d. $12x^4 - 10x^2 + 7$

22 [Foundational] • Target: Derivatives of Polynomials & Tangents

Find the gradient of the curve $y = x^3 - 4x + 1$ at the point where $x = 2$.

a. 8

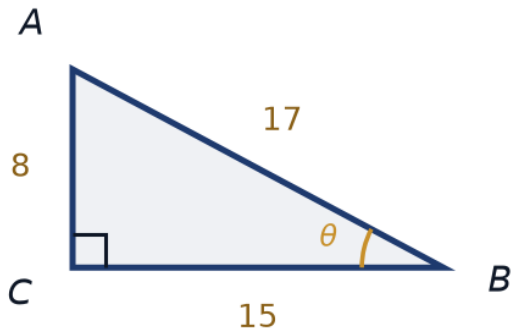
b. 12

c. 4

d. 0

23 [Foundational] • Target: Derivatives of Polynomials & Tangents

Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point $(3, 2)$.



a. $y = 3x - 7$

b. $y = 3x - 2$

c. $y = x - 1$

d. $y = 2x - 4$

24 [Foundational] • Target: Stationary Points & Optimization

Find the stationary points of the curve $y = 2x^3 - 9x^2 + 12x + 1$ and determine their nature.

a. Local max at $(1, 6)$, local min at $(2, 5)$

b. Local min at $(1, 6)$, local max at $(2, 5)$

c. Local max at $(2, 5)$, inflection at $(1, 6)$

d. Local min at $(0, 1)$, local max at $(2, 5)$

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25 [Foundational] • Target: Stationary Points & Optimization

A metal wire of length 40 cm is bent into a rectangle of dimensions x by y. What value of x maximizes the enclosed area?

a. $x = 10$ cm

b. $x = 20$ cm

c. $x = 5$ cm

d. $x = 15$ cm

26 [Foundational] • Target: Stationary Points & Optimization

For what range of values of x is the function $f(x) = x^3 - 6x^2 + 9x - 4$ strictly increasing?

a. $x < 1$ or $x > 3$

b. $1 < x < 3$

c. $x > 2$

d. $x < 2$

27 [Foundational] • Target: Indefinite & Definite Integrals

Find $\int (6x^2 - 4x + 5)\,dx$.

a. $2x^3 - 2x^2 + 5x + C$

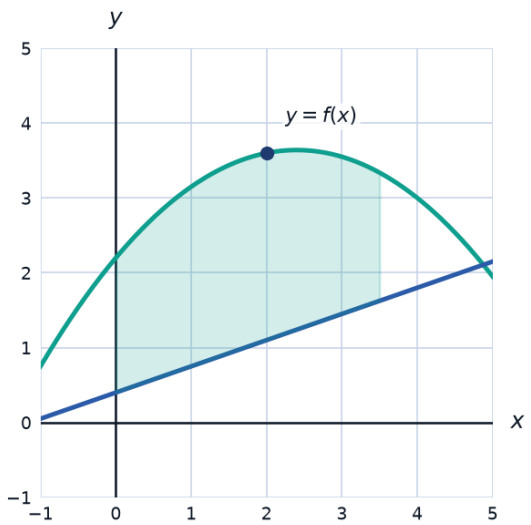
b. $3x^3 - 2x^2 + 5x + C$

c. $2x^3 - 4x^2 + 5x + C$

d. $12x - 4 + C$

28 [Foundational] • Target: Indefinite & Definite Integrals

Evaluate the definite integral $\int_1^3 (3x^2 - 2x)\,dx$.



a. 18

b. 24

c. 16

d. 20

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Indefinite & Definite Integrals**

A curve passes through $(1, 4)$ and has gradient function $\frac{dy}{dx} = 4x - 1$. Find the equation of the curve.

a. $y = 2x^2 - x + 3$

b. $y = 2x^2 - x + 4$

c. $y = 4x^2 - x + 1$

d. $y = 2x^2 - x + 1$

30 **[Intermediate]** • **Target: Area Under Curves & Exam-Style Modeling**

Find the area enclosed by the curve $y = 4 - x^2$ and the x-axis.

a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 8

d. $\frac{64}{3}$

31 **[Intermediate]** • **Target: Area Under Curves & Exam-Style Modeling**

Find the area of the region bounded by the curve $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

32 **[Intermediate]** • **Target: Area Under Curves & Exam-Style Modeling**

A particle moves along a straight line such that its velocity at time t is $v(t) = 3t^2 - 12t + 9$. Find the total displacement between $t = 0$ and $t = 3$.

a. 0

b. 9

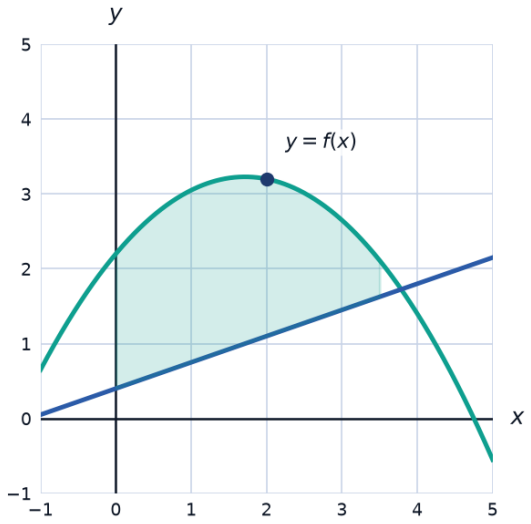
c. 18

d. 4.5

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33 [Intermediate] • Target: Derivatives of Polynomials & Tangents

Find the area enclosed between the curve $y = 3x$, the x-axis, and the lines $x = 0$ and $x = 2$.



a. 11

b. 5

c. 12

d. 6

34 [Intermediate] • Target: Stationary Points & Optimization

Find the area enclosed between the curve $y = 6x$, the x-axis, and the lines $x = 0$ and $x = 6$.

a. 107

b. 113

c. 108

d. 216

35 [Intermediate] • Target: Indefinite & Definite Integrals

Find the area enclosed between the curve $y = 6x$, the x-axis, and the lines $x = 0$ and $x = 4$.

a. 47

b. 53

c. 96

d. 48

36 [Intermediate] • Target: Area Under Curves & Exam-Style Modeling

Find the area enclosed between the curve $y = 4x$, the x-axis, and the lines $x = 0$ and $x = 4$.

a. 32

b. 64

c. 37

d. 31

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37 [Intermediate] • Target: Derivatives of Polynomials & Tangents

Find the area enclosed between the curve $y = 2x$, the x-axis, and the lines $x = 0$ and $x = 4$.

a. 15

b. 21

c. 32

d. 16

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Stationary Points & Optimization • Cross-Lesson: 5.01 Trigonometry

Find the area enclosed between the curve $y = 2x$, the x-axis, and the lines $x = 0$ and $x = 6$.

a. 72

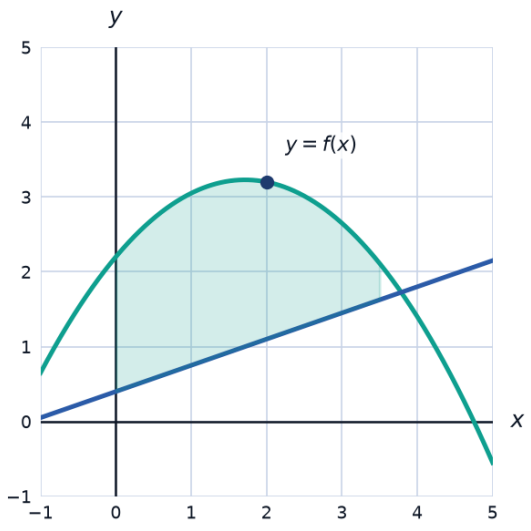
b. 35

c. 41

d. 36

39 [Challenging] • Target: Indefinite & Definite Integrals • Cross-Lesson: 7.01 Statistics & Probability

Find the area enclosed between the curve $y = 2x$, the x-axis, and the lines $x = 0$ and $x = 2$.



a. 4

b. 3

c. 8

d. 9

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40 [Challenging] • Target: Area Under Curves & Exam-Style Modeling • Cross-Lesson: 8.01 Functions: Composite & Inverse

Find the area enclosed between the curve $y = 2x$, the x-axis, and the lines $x = 0$ and $x = 3$.

a. 8

b. 18

c. 14

d. 9

41 [Challenging] • Target: Derivatives of Polynomials & Tangents • Cross-Lesson: 9.01 Circular Measure

Find the x-coordinate of the stationary point of $y = 4x^2 + 10x - 3$.

a. -1/4

b. 5/4

c. -5/4

d. -9/4

42 [Challenging] • Target: Stationary Points & Optimization • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Find the x-coordinate of the stationary point of $y = 3(x + 1)^2 - 2$.

a. 1

b. 0

c. -1

d. -2

43 [Challenging] • Target: Indefinite & Definite Integrals • Cross-Lesson: 10.01 Series & Binomial Theorem

Find the x-coordinate of the stationary point of $y = x^2 + x + 8$.

a. -1/2

b. 1/2

c. -3/2

d. 5/2

44 [Challenging] • Target: Area Under Curves & Exam-Style Modeling • Cross-Lesson: 11.01 Kinematics

Find the x-coordinate of the stationary point of $y = (x - 5)^2 - 5$.

a. -5

b. 5

c. 4

d. 6

45 [Challenging] • Target: Derivatives of Polynomials & Tangents • Cross-Lesson: 12.01 Forces & Equilibrium

Find the x-coordinate of the stationary point of $y = (x - 4)^2 + 4$.

a. -4

b. 4

c. 5

d. 3

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